Scott Wilson

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SUMMARY

I am a computational biologist, who has worked across various high dimensional datasets to extract signal from noise. I recently completed my PhD in neuroscience at the University of Cambridge. Eager to pursue a career which leads to macro-level, positive outcomes across a changing world.

RESEARCH EXPERIENCE

2021–present

PhD & Postdoctoral researcher
MRC Laboratory of Molecular Biology, Cambridge — Cardona group

- Analysed the complete larval connectome (∼3,000 neurons, ∼500,000 synapses), the world's most complex brain network at the time.
- Generated and analysed single-cell and spatial transcriptomic data on how learning reshapes gene expression. Likewise utilised CRISPR-Cas genetic engineering.
- Built reproducible computational pipelines for large imaging and omics datasets, and their analysis.

2020–2021

MSci research thesis — Computational Biology
The Francis Crick Institute — Luscombe lab

- *A computational analysis of proteomic and epigenomic perturbations in a patient-derived ALS model system*.
- Participated in the Crick Data Challenge 2020, with my team winning the prize ('Leveraging deep learning to predict gene essentiality from genome-wide CRISPR screens').

2019–2020

Research assistant
Institute of Healthy Ageing, UCL — Partridge lab

- Genetic screens (RNAi, Gal4–UAS) for pathways modulating neurodegenerative phenotypes; assessed via lifespan assays, western blots.

2023–present

Working group — "Assessing Representation in Minds and Artificial Systems"
Santa Fe Institute

- Selected (1 of 12) to work with Melanie Mitchell, David Krakauer, and others on how minds and machines represent the world.

PUBLICATIONS

in press — **Wilson SR**, et al. The structure of developmental noise in a complete connectome. first author

in press — **Wilson SR**, et al. Spatial transcriptomics of the larval mushroom body reveals intrinsic pretuning of memory-circuit neurons. first author

in press — **Wilson SR**, et al. Towards the panconnectome. first author

2026 — Mace C, O'Sullivan F, **Wilson SR**. Reductionism in engram neuroscience. *European Journal of Neuroscience*. doi:10.1111/ejn.70497 co-first

2023 — Ferreira Castro A, **Wilson S**, Cardona A. Evaluating traces of Hebbian plasticity in the *Drosophila* antennal lobe. *PNAS* **120**. doi:10.1073/pnas.2315790120 co-author

EDUCATION

2021–2026

PhD, Neuroscience
University of Cambridge — MRC Laboratory of Molecular Biology

- Thesis: *'On the Individuality of a Brain: Connectomic and Transcriptomic Plasticity in the Drosophila Larva'*; three first-author papers in press.

2017–2021

MSci Biological Sciences (Genetics) — First Class Honours
University College London

- Recognised on the Dean's List, as top 5% of students in faculty.
- Awarded prize for the best undergraduate dissertation in a cohort of around 200: *'The Epigenetic Inheritance of Longevity'*.
- Modules incl. Neural Computation, Advanced Computational Biology, Advanced Human Genetics.

TECHNICAL SKILLS

Languages Python, R, Git, Command line
Data Single-cell & spatial RNA-seq, bulk transcriptomics & proteomics, connectomics
Methods Statistical modelling & hypothesis testing, deep learning, dimensionality reduction, network analysis, scientific visualisation
Wet lab CRISPR-Cas9 engineering, optogenetics, electron & confocal microscopy

AWARDS & ACTIVITIES

- British Biology Olympiad — Gold, 2017.
- Harold & Olga Fox Prize, UCL (Highest-graded BSc dissertation in my faculty) — 2020.
- Crick Data Challenge 2020 (winning team).
- Student, Neuromatch Academy (Computational Neuroscience), 2021.
- Tutor, Neuromatch Academy (Computational Neuroscience), 2023.
- President, Wolfson College Boat Club, Cambridge — 2023–25.
- International finalist for the UK on *Super Brain* (最强大脑), Seasons 11 (2024) & 12 (2025).